

Case Study: Camera Installation Decision for Vending Machines

Background

You are a vending machine operator managing a network of 3,000 machines across school buildings in Taipei City. The company has observed occasional vandalism of vending machines and is considering installing surveillance cameras at high-risk locations to deter such damage.

Each vending machine location is described by environmental and contextual attributes, including building type, whether it's placed indoors, and whether a guard is stationed nearby. The goal is to predict whether a given vending machine should be equipped with a camera.

The dataset includes the following variables:

- **Building Type:** One of the following categories: dorm, classroom, gym, or office.
- **Indoor:** Indicator of whether the vending machine is placed indoors.
- **Distance to Nearest Convenience Store (meters):** Ranges from 0 to 500.
- **Weekend:** Binary indicator for whether the day is a weekend.
- **Club Activity Day:** Indicates whether club activities are held near the machine.

An additional dataset indicates whether each vending machine has been vandalized in the past year.

Question 1: Descriptive Analysis

Using the historical dataset of 3,000 machines:

- Provide summary statistics for each feature, separated by **Broken** = 0 and **Broken** = 1. What variables show the most noticeable differences between the two groups?
- Visualize the distribution of variables with respect to whether it is broken.

Question 2: Classification Modeling

- Run a **logistic regression model (logit)** to estimate the probability that a vending machine will be vandalized (i.e., **Broken** = 1) using the provided features. Report the estimated coefficients. Which variables have the strongest influence on the probability of being broken?

The model takes the following form:

$$\begin{aligned} \text{logit}(\Pr(\text{Broken}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{Indoor}_i + \beta_2 \cdot \text{DistanceToStore}_i \\ & + \beta_3 \cdot \text{Weekend}_i + \beta_4 \cdot \text{ClubActivityDay}_i \\ & + \sum_{k=1}^K \gamma_k \cdot \text{BuildingType}_{ik} \end{aligned} \quad (1)$$

- (b) Report classification performance using confusion matrix, accuracy, precision, and recall, and plot the AUC Curve.
- (c) Repeat 2-1 and 2-2 with random forest. Compare the confusion matrix, ROC, and AUC.
- (d) Which features are most influential in determining the need for surveillance?

Question 3: Business Recommendation

Your model predicts which vending machines are at higher risk of vandalism and therefore may benefit from camera installation. However, the company must also consider the economic tradeoff between installing a camera and the potential cost of damage if a machine is left unprotected.

- (a) Which machine should we install camera? Assume that installing a camera costs \$2,000, and the potential loss from vandalism is \$10,000. How many machines should we install a camera?
- (b) Summarize the common characteristics of these high-risk machines. Are there patterns in location, indoor/outdoor placement?
- (c) Compare your model-based camera deployment strategy with a simple human rule that installs cameras only on outdoor machines. Does your predictive model provide a better cost-benefit outcome than the human rule?
- (d) Sensitivity analysis with different assumptions on camera effectiveness. Vary the effectiveness from 0% to 100%, and calculate the total expected loss under the model-based strategy.